AI / ML PORTFOLIO ARTIFACT 02

Training a large language model

How data becomes a deployed system, and where resources go

Rahul Niroop November 8, 2026 Published examples from 2023-2025

01 FROM DATA TO DEPLOYMENT

01 Collect data — D T
Gather text and code with documented rights and provenance.
Coverage and licensing shape quality, risk, and acquisition cost.

02 Prepare and tokenize — D C
Clean text, protect privacy, split test data, and create tokens.
Tokens are text pieces. Keep evaluation examples out of training.

03 Design and pilot — C T
Choose model size and architecture. Pilot small runs before scaling.
Fit useful model capacity and training data to a compute budget.

04 Pretrain — D C E
Predict tokens, measure error, and update the learned model weights.
Gradients guide updates. Many GPUs repeat this across batches.

05 Post-train — D C T
Teach instructions with examples, then refine responses with feedback.
Human ratings and checked synthetic examples guide model behavior.

06 Evaluate and improve — D T
Test task accuracy, factuality, bias, privacy, and misuse resistance.
Use separate tests. Failed checks can send work back to earlier stages.

07 Deploy and monitor — C E T
Optimize and serve the model. Monitor response speed and reliability.
Generation uses ongoing compute. Ordinary prompts do not update weights.

Letter badges link stages to typical resource emphasis, not exclusive use or measured budget shares.
Evaluation can return work to data preparation or model design. Skilled people support every stage. [1-3, 6, 8]

02 RESOURCE KEY

D Data
Text, code, storage, curation
Relevant, diverse examples support coverage. Licensing, cleaning, and annotation cost money. More raw data alone does not assure quality.

C Compute
Accelerators, memory, networks
GPUs or TPUs perform the training math. Fast networks share work. Memory holds weights and other state needed to update the model.

E Energy
Electricity and cooling
Power sustains the cluster. Hardware efficiency and utilization affect use. Carbon also depends on the electricity source and accounting method.

T Time
Run duration and iteration
Data preparation, experiments, and feedback extend the project. Parallel hardware can shorten a run, subject to memory and network limits.

03 WHAT DEVELOPERS HAVE DISCLOSED

GPU-hours shown on a common scale (millions). Hardware and scope differ.
These bars show reported usage, not comparable speed, efficiency, or model quality.

Llama 2 70B
2.0T pretraining tokens
A100 80GB / Pretraining
1.720M GPU-hours [4]
688 MWh power-capacity proxy
Dollar training budget not stated in the cited card.

Llama 3.1 405B
15T+ pretraining tokens
H100 80GB / Training, as reported
30.840M GPU-hours [5]
21.59 GWh power-capacity proxy
Dollar training budget not stated in the cited card.

DeepSeek-V3
14.8T pretraining tokens
H800 / Official training total
2.788M GPU-hours [6]
$5.576M at assumed $2/GPU-hour
Excludes prior research and ablations. Not a total project budget.

GPT-4 (2023): training hardware, compute, model size, and dataset construction
are withheld in its technical report. A verified full training dollar budget is not provided there. [7]

04 READ THE UNITS BEFORE THE PRICE TAG

Time is not the same as GPU-hours

DeepSeek-V3 pretraining: 2.664M GPU-hours [6]

2,664,000 / (2,048 GPUs x 24) = 54.2 days

Calculated at full occupancy. The report says less than two months. Research and preparation add project time.

Energy depends on the boundary

Llama 3.1 405B: 30.84M hours x 0.700 kW [5]

= 21.588M kWh = 21.59 GWh

Capacity-based proxy, not metered electricity. Cooling and other equipment depend on the accounting boundary.

B = billion parameters. T = trillion tokens in model data labels. A GPU-hour = one GPU for one hour.
GPU = graphics processing unit. TPU = tensor processing unit. D, C, E, T badges use the resource key above.
Sources: [1] Wolfram [2] IBM [3] Ouyang et al. [4] Llama 2 [5] Llama 3.1 [6] DeepSeek-V3 [7] GPT-4 [8] Hoffmann et al.
Full sources, accessible text, calculations, and explanatory document